Redbird Flight Simulations
Qualification and Approval Guide (QAG)

LD 1000, SD 1000, and FMX 1000 Advanced Aviation Training Devices
Amended Qualification and Approval Guide
VERSION 4.2
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Section 1: List of Configurations

Currently Approved:

1. **Configuration: Generic Complex Single Engine-** Land

2. **Configuration: C172-XX-S001-** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, High Wing

3. **Configuration: C172-XX-G001-** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, High Wing

4. **Configuration: C182-XX-S001-** Single Engine, Land, Fixed Gear, Constant Speed Propeller, High Wing

5. **Configuration: C182-XX-G001-** Single Engine, Land, Fixed Gear, Constant Speed Propeller, High Wing

6. **Configuration: PA28A-XX-S001-** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, Low Wing

7. **Configuration: PA28R-XX-S001-** Single Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

8. **Configuration: SR20-XX-G001-** Single Engine, Land, Fixed Gear, Constant Speed Propeller, Low Wing

9. **Configuration: SR22-XX-G001-** Single Engine, Land, Fixed Gear, Constant Speed Propeller, Low Wing

10. **Configuration: BE58-XX-S001-** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

11. **Configuration: PA34-XX-S001-** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

12. **Configuration: PA44-XX-S001-** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

Currently Approved:

13. **Configuration: C172-FC-G003-** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, High Wing

14. **Configuration: C182-FC-G003-** Single Engine, Land, Fixed Gear, Constant Speed Propeller, High Wing

15. **Configuration: PA28A-XX-G001-** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, Low Wing

16. **Configuration: PA28B-XX-S001-** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, Low Wing

17. **Configuration: PA28R-XX-G001-** Single Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

18. **Configuration: DA20-XX-S007-** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, Low Wing

19. **Configuration: DA40-FC-G003-** Single Engine, Land, Fixed Gear, Constant Speed Propeller, Low Wing

20. **Configuration: SR22-FC-G004-** Single Engine, Land, Fixed Gear, Constant Speed Propeller, Low Wing

21. **Configuration: BE55-XX-S008-** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

22. **Configuration: BE58-FC-G003-** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

23. **Configuration: BE76-XX-S006-** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

24. **Configuration: PA30-XX-S009-** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

25. **Configuration: GA7-XX-S009-** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

26. **Configuration: DA42-FC-G003-** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

Statements of Compatibility of Software and Hardware
AC61-136, Appendix 2, 4. b.:

The device and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.

Microsoft Windows XP or 7 (Operating System)
"This is to certify that Microsoft Corporation, the owner and developer of the Windows 7 operating system, has evaluated that their operating system works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software."

Microsoft ESP or FSX (Simulation Engine)
"This is to certify that Microsoft Corporation, the owner and developer of the ESP and FSX Simulator Engine, has evaluated that their software application works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software."

Lockheed Martin Prepar3D (Simulation Engine)
"This is to certify that Lockheed Martin, the owner and developer of the Prepar3D Simulator Engine, has evaluated that their software application works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software."

Mindstar Aviation Redbird Virtual Instrument Suite
"This is to certify that Mindstar Aviation, the owner and developer of the Redbird Redbird Virtual Instrument Suite, including the Redbird 1000, Redbird 430/530, Redbird 140 Autopilot, Redbird 55 Autopilot, Redbird Entegra and Redbird Flight Simulations, the hardware manufacture and component integrator, have demonstrated that the Redbird Virtual Instrument Suite software package is fully compatible with the Redbird Flight Simulations LD, SD, FMX 1000 and all configurations of that model. Mindstar Aviation and Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations."

Statements of Compatibility of Software and Hardware
AC61-136, Appendix 2, 4. b. (continued):

The device and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.

Flight 1 Compass

"This is to certify that Flight 1, the owner and developer of the Compass and Redbird Flight Simulations, the hardware manufacture and component integrator, have demonstrated that the Compass software is fully compatible with the Redbird Flight Simulations LD, SD, FMX 1000 and all configurations of that model. Flight 1 and Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations."

Section 2: Description of Hardware and Software Components

All Configurations Include:

Qty	Type	Manufacturer	Name	Description/Function	Configuration
1	Software	Microsoft	Windows XP or 7	Operating system. (Software version depending on serial number)	All
1	Software	Microsoft	ESP or FSX	Simulation engine. (Software version depending on serial number)	All
1	Software	Lockheed Martin	Prepar3D	Simulation engine. (Software version depending on serial number)	All
1	Software	Redbird Flight Simulations	RB Sim	Simulation control and component integration.	All
1	Software	Redbird Flight Simulations	FMX Control	Motion system control and component integration.	FMX only
1	Software	Redbird Flight Simulations	Instructors Station	Environmental, location and failure controls with map, track and glideslope display.	All
1	Software	Mindstar Aviation	Redbird 1000	Virtual flight instruments, GPS, radios, gauges, indicators, alerts, misc. instruments and logic controls for simulated systems.	See Configuration
1	Software	Mindstar Aviation	Redbird 430/ 530	Virtual GPS and radios	See Configuration
1	Software	Mindstar Aviation	Redbird Entegra	Virtual flight instruments, radios, gauges, indicators, alerts, misc. instruments and logic controls for simulated systems.	See Configuration
1	Software	Mindstar Aviation	Redbird Autopilot	Virtual Autopilot (Redbird 140, 55, or 700)	See Configuration
1	Software	Flight 1	Compass	Virtual Compass	All
1	Software	Redbird Flight Simulations	Analog Gauges	Virtual Airspeed, Attitude, Altimeter, Turn and Bank, Heading, HSI, VSI, RMI, CDI, and ADF Gauges	See Configuration
1	Software	Redbird Flight Simulations	Miscellaneous Gauges	Virtual Miscellaneous Gauges	All
1	Hardware	Redbird Flight Simulations	Simulation Computer	Host computer for flight simulation engine, simulation control software, airplane systems and instruments.	All
1	Hardware	Redbird Flight Simulations	Motion Control Computer	Host computer for FMX Control.	FMX only
1	Hardware	Industry Standard	Instructors Station Computer	Host computer for Instructors Station.	All
1	Hardware	Redbird Flight Simulations	Motion Platform	Gimbaled, steel motion platform with movement in pitch, roll and yaw. Includes all motors, sensors and safety controls.	FMX only- See Figure 46, 47,48
1	Hardware	Redbird Flight Simulations	Cockpit Enclosure	Cockpit enclosure to exclude distractions.	All- See Figure 46, 47,48
6	Hardware	22” Industry Standard	LCD	Flat Panel displays for exterior views.	All
2	Hardware	19” Industry Standard	LCD	Flat Panel displays for virtual instruments.	All
1	Hardware	Redbird Flight Simulations	Yoke, Side Yoke, Center Stick	Pitch and roll controller with switches and buttons for airplane systems operation. As required for each configuration.	All- See Configuration
1	Hardware	Redbird Flight Simulations	Throttle Quadrant	As required for each configuration.	All- See Configuration
2	Hardware	Redbird Flight Simulations	Rudder Pedals	Rudder control pedals with toe brakes.	All
1	Hardware	Redbird Flight Simulations	Switch Panel	Lower switch panel with airplane configuration and systems controls.	All
1	Hardware	Redbird Flight Simulations	Instrument Controls Overlay	Flight instruments, radios, airplane configuration and systems controls as required for each configuration.	All- See Configuration

Approved Individual Configurations Include:

Generic Complex Single Engine Land: Approved Configuration
Yoke: Standard yoke. See Figure— 27
Throttle Quadrant: Single Engine Complex Vernier. See Figure— 31
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530,Redbird 140
Autopilot. See Figure— 1

C172-XX-S001: Approved Configuration
Yoke: Standard yoke. See Figure— 27
Throttle Quadrant: Single Engine Vernier. See Figure— 30
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140
Autopilot. See Figure— 2

C172-XX-G001: Approved Configuration
Yoke: Standard yoke. See Figure— 27
Throttle Quadrant: Single Engine Vernier. See Figure— 30
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Redbird 140
Autopilot. See Figure—3

C182-XX-S001: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Single Engine Complex Vernier. See Figure— 31
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530,Redbird 140
Autopilot. See Figure— 5

C182-XX-G001: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Single Engine Complex Vernier. See Figure— 31
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Redbird 140
Autopilot. See Figure— 6

PA28A-XX-S001: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 32
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140
Autopilot. See Figure— 8

PA28R-XX-S001: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Single Engine Complex “Boat Style”. See Figure— 33
Virtual Instrument Controls Overlay: Traditional Round Gauges, Dual Redbird 430, Redbird 55
Autopilot. See Figure— 11

SR20-XX-G001: Approved Configuration
Yoke: Side Yoke. See Figure— 28
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 32
Virtual Instrument Controls Overlay: Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot. See Figure— 15

SR22-XX-G001: Approved Configuration
Yoke: Side Yoke. See Figure— 28
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 32
Virtual Instrument Controls Overlay: Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot. See Figure— 16

BE58-XX-S001: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Twin Engine “Boat Style”. See Figure— 34
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 19

PA34-XX-S001: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Twin Engine “Boat Style”. See Figure— 34
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 23

PA44-XX-S001: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Twin Engine “Boat Style”. See Figure— 34
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 24

C172-FC-G003: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Single Engine Vernier. See Figure— 30
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Integrated Redbird 700 Autopilot. See Figure— 4

C182-FC-G003: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Single Engine Complex Vernier. See Figure— 31
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Integrated Redbird 700 Autopilot. See Figure— 7

PA28A-XX-G001: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 32
Virtual Instrument Controls Overlay: Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot. See Figure— 9

PA28B-XX-S001: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 32
Virtual Instrument Controls Overlay: Traditional Round

Gauges, Dual Redbird 430, Redbird 55 Autopilot. See Figure— 10

PA28R-XX-G001: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Single Engine Complex “Boat Style”. See Figure— 33
Virtual Instrument Controls Overlay: Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot. See Figure—12

DA20-XX-S007: Approved Configuration
Yoke: Center Stick. See Figure— 29
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 32
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430, Redbird 140 Autopilot. See Figure— 13

DA40-FC-G003: Approved Configuration
Yoke: Center Yoke. See Figure— 29
Throttle Quadrant: Single Engine Complex “Boat Style”. See Figure— 33
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Integrated Redbird 700 Autopilot. See Figure— 14

SR22-FC-G003: Approved Configuration
Yoke: Side Yoke. See Figure— 28
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 32
Virtual Instrument Controls Overlay: Glass Cockpit Redbird Perspective PFD & MFD with Redbird FMS Keypad and Redbird Perspective Autopilot. See Figure— 17

BE55-XX-S008: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Twin Engine Reverse “Boat Style”. See Figure— 35
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 18

BE58-FC-G003: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Twin Engine “Boat Style”. See Figure— 34
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Integrated Redbird 700 Autopilot. See Figure— 20

BE76-XX-S003: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Twin Engine “Boat Style”. See Figure— 34
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 21

PA30-XX-S009: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Twin Engine “Boat Style”. See Figure— 34
Virtual Instrument Controls Overlay: Traditional Round

Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 22

DA42-FC-G003: Approved Configuration
Yoke: Center Stick. See Figure— 29
Throttle Quadrant: Twin Engine FADEC "Boat Style". See Figure— 36
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Integrated Redbird 700 Autopilot. See Figure— 25

GA7-XX-S009: Approved Configuration
Yoke: Standard Yoke. See Figure— 27
Throttle Quadrant: Twin Engine "Boat Style". See Figure— 34
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 26

Section 3: Statements of Compliance

All Configurations, as noted, meet AC 61-136, Appendix 2, Paragraph 8 BATD Design Criteria:

a. Controls:

(1) Meets Requirements: Physical flight and aircraft system controls are recognizable as to their function and how they are manipulated solely from their appearance. There is no keyboard, mouse, or gaming joystick inside the LD/SD/FMX 1000 Cockpit. See Figures— 44, 45

(2) Meets Requirements: Virtual controls are limited to the initialization of the device and instructor input. Avionics are virtually displayed but are operated with physical buttons and knobs that closely represent the configuration of the aircraft. See Figures— 37, 38, 44, 45

(3) Meets Requirements: No keyboard or mouse is used to set or position any feature of the device. The required and additional equipment is operated in the same manner in which it would be operated in the represent aircraft.

(4) Meets Requirements: The arrangement, appearance, and operation of controls, instruments, and switches closely represent the aircraft, including: master/battery, magnetos, alternators, fuel boost pumps, avionics master, pitot heat, and required lights. See Figures— 37, 38, 44, 45

(5) Meets Requirements: During start-up of the device the PC configuration and file inventory is examined to ensure only tested software is installed. In the event additional files are found, an initialization error will be displayed and logging of the flight training data to the student key will not be allowed.

b. Control Requirements:

(1)(a): Meets Requirements: See individual configuration.
(1)(b): Meets Requirements: See Figure— 42
(1)(c): Meets Requirements: See Figure— 30, 31, 32, 33, 34, 35, 36
(1)(d): Meets Requirements: See individual configuration.
(1)(e): Meets Requirements: See individual configuration.

c. Control Inputs:

(1) Meets Requirements: The transport delay for all control inputs is 300 milliseconds or less. The required transport delay is tested at system start-up by internal start/stop timers. Redbird Flight Simulations is the manufacturer of all control inputs and uses USB 2.0 which exceeds this requirement.

(2) Meets Requirements: Each control input is tested at training session start-up. If any device fails to meet the requirements of Appendix 2, Paragraph 8, Section c, item (1) a warning message is displayed and the device will not start a training session.

d. Display Requirements:

(1) Meets Requirements: See individual configuration.

(2) Meets Requirements: All instrument displays included in each individual configuration are visible during all flight operations. The displays are updated at rate that insures that the image of the instrument does not appear to: be out of focus, illegible, jump, step, jagged, nor is it distracting.

(3) Meets Requirements: Display updates exceed 60Hz. All replicated instruments display changes equal or greater than the values required by this item.

(4) Meets Requirements: Displays reflect the dynamic behavior of an actual aircraft in flight.

e. Flight Dynamics Requirements:

(1) Meets Requirements: All simulated aircraft flight dynamics are comparable to the way an actual aircraft performs and handles.

(2) Meets Requirements: All simulated aircraft performance parameters are comparable to the aircraft being represented.

(3) Meets Requirements: Vertical lift component changes as a function of bank, comparable to the aircraft being represented.

(4) Meets Requirements: Changes in aircraft configuration are accompanied by changes in flight dynamics, comparable to the aircraft being represented.

(5) Meets Requirements: The handling and performance qualities of the simulated aircraft are affected by the presence and intensity of wind and turbulence in manner comparable to the aircraft being represented.

f. Instructional Management Requirements:

(1) Meets Requirements: The instructor has the ability to pause the system at any time using either the buttons on the instructor's console or the buttons on the instrument panel. See Figure— 40, 41

(2) Meets Requirements: The instructor as the ability to manipulate the parameters listed in this item independently of the simulation.

(3) Meets Requirements: The system records both the horizontal and vertical track of the aircraft movement and this can be view in real-time or at a later time.

(4) Meets Requirements: The instructor is able to disable instruments prior to or during a training session and simulate failure of the instruments without stopping or freezing the

simulation to affect the failure.

(5) Meets Requirements: The device includes a navigational data base that covers all of North America and is based on procedures as published per 14 CFR part 97.

All configurations, as noted meet AC 61-136, Appendix 3, Paragraph 3 AATD Design Criteria:

a.

(1) Meets Requirements: The device enclosure and cockpit design and instrument panel arrangement represent a generic aircraft commonly used by flight training operations. See Figure— 44, 45, 46, 47, 48

(2) Meets Requirements: Cockpit knobs/system controls/switches/switch panels are realistic in size and design appropriate to each intended function, in the proper position and distance from the pilot's seated position, and representative of the category and class of the aircraft being represented. See individual configuration.

(3) Meets Requirements: Primary flight and navigation instruments are life size and exhibit neither stepping nor excessive transport delay. They are arranged so as to observe trends and provide a realistic scan pattern. See individual configuration.

(4) Meets Requirements: See individual configuration.

(5) Meets Requirements: See individual configuration.

(6) Meets Requirements: See individual configuration.

(7) Meets Requirements: Manual pitch trim is provided via a trim wheel on all configurations. Individual configurations may have electric trim as well. See individual configuration. See Figure— 39

(8) Meets Requirements: All visual cues for both day and night, VFR and IFR are provided. See Figure— 43

(9) Meets Requirements: Instructors station is run from a "tablet" style or laptop PC from inside or outside the device. The instructor can oversee the pilot's horizontal and vertical flight profiles in real time and space. The instructor has the ability to complete all functions listed in this item without interrupting the flight. See Figure— 40

(10) Cockpit seat is fully adjustable.

Approved Configuration: Generic Complex Single Engine—Land

This configuration represents a generic, high wing, complex, single engine land, airplane.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 31
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 1

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the

requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels.
See Figure— 1

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated.
See Figure— 1

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot.
See Figure— 1

(5) Redbird 430/530. See Figure— 1

(6) Redbird 140 Autopilot. See Figure— 1

(7) Manual pitch trim with indicator. See Figures— 39, 1

Approved Configuration: C172-XX-S001

This configuration represents a single engine land, fixed gear, fixed pitch propeller, high wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has mixture and throttle controls. See Figure— 30
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 2

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 2
Sensitive altimeter that meets the requirements

listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item. Gyroscopic rate-of-turn indicator that meets the requirements listed in this item. Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item. Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 2

(3) Life sized traditional “round” gages and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 2

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 2

(5) Redbird 430/530. See Figure— 2

(6) Redbird 140 Autopilot. See Figure— 2

(7) Manual pitch trim with indicator. See Figure— 39, 2

Approved Configuration: C172-XX-G001

This configuration represents a single engine land, fixed gear, fixed pitch propeller, high wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and a Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has mixture and throttle controls. See Figure— 30
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.
This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 3

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 3
Sensitive altimeter that meets the requirements

listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item. Gyroscopic rate-of-turn indicator that meets the requirements listed in this item. Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item. Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 3
(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 3
(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Redbird 140 Autopilot. See Figure— 3
(5) Glass Cockpit Redbird 1000 PFD & MFD . See Figure— 3
(6) Redbird 140 Autopilot. See Figure— 3
(7) Manual pitch trim with indicator. See Figure— 39, 3

Approved Configuration: C182-XX-S001

This configuration represents a single engine land, fixed gear, constant speed propeller, high wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 31
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 5

d. Display Requirements:

(1) This configuration has instruments and

indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 5
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 5

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 5

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 5

(5) Redbird 430/530. See Figure— 5

(6) Redbird 140 Autopilot. See Figure— 5

(7) Manual pitch trim with indicator. See Figure— 39, 5

Approved Configuration: C182-XX-G001

This configuration represents a single engine land, fixed gear, constant speed propeller, high wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and a Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 30
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 6

d. Display Requirements:

(1) This configuration has instruments and

indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 6
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 6

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 6

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Redbird 140 Autopilot. See Figure— 6

(5) Glass Cockpit Redbird 1000 PFD & MFD . See Figure— 6

(6) Redbird 140 Autopilot. See Figure— 6

(7) Manual pitch trim with indicator. See Figure— 39, 6

Approved Configuration: PA28A-XX-S001

This configuration represents a single engine land, fixed gear, fixed pitch propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has mixture and throttle controls. See Figure— 32
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 8

d. Display Requirements:

(1) This configuration has instruments and

indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 8
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 8

(3) Life sized traditional “round” gages and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 8

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 8

(5) Redbird 430/530. See Figure— 8

(6) Redbird 140 Autopilot. See Figure— 8

(7) Manual pitch trim with indicator. See Figure— 39, 8

Approved Configuration: PA28R-XX-S001

This configuration represents a single engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird Dual 430, and Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 33
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Dual Redbird 430).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 11

d. Display Requirements:

(1) This configuration has instruments and

indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 11
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item(Dual Redbird 430).
Navigation radio that meets the requirements listed in this item (Dual Redbird 430).

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 11

(3) Life sized traditional “round” gauges and Redbird Dual 430 GPS and Navigation radios representative of aircraft being simulated See Figure— 11

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 11

(5) Redbird 430/530. See Figure— 11

(6) Redbird 55 Autopilot. See Figure— 11

(7) Manual pitch trim with indicator. See Figure— 39, 11

Approved Configuration: SR20-XX-G001

This configuration represents a single engine land, fixed gear, constant speed propeller, low wing, airplane with Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering side yoke. See Figure— 28
(1)(d) This configuration has mixture and throttle controls. See Figure— 32
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Dual Redbird 430).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 15

d. Display Requirements:

(1) This configuration has instruments and

indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird Entegra PFD & MFD with standby instruments. See Figure-15
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Dual Redbird 430)
Navigation radio that meets the requirements listed in this item. (Dual Redbird 430)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird Entegra PFD & MFD with standby instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 15

(3) Life sized Glass Cockpit Redbird Entegra PFD & MFD and dual Redbird 430 GPS and Navigation radios representative of aircraft being simulated See Figure— 15

(4) Dual Redbird 430, Transponder, Audio Panel, and Redbird 55 Autopilot. See Figure— 15

(5) Dual Redbird 430 . See Figure— 15

(6) Redbird 55 Autopilot. See Figure— 15

(7) Manual and electric pitch trim with indicator. See Figure— 39, 28, 15

Approved Configuration: SR22-XX-G001

This configuration represents a single engine land, fixed gear, constant speed propeller, low wing, airplane with Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering side yoke. See Figure— 28
(1)(d) This configuration has mixture and throttle controls. See Figure— 32
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Dual Redbird 430).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 16

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:

Glass Cockpit Redbird Entegra PFD & MFD with standby instruments. See Figure- 16
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Dual Redbird 430)
Navigation radio that meets the requirements listed in this item. (Dual Redbird 430)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird Entegra PFD & MFD with standby instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 16

(3) Life sized Glass Cockpit Redbird Entegra PFD & MFD and dual Redbird 430 GPS and Navigation radios representative of aircraft being simulated See Figure— 16

(4) Dual Redbird 430, Transponder, Audio Panel, and Redbird 55 Autopilot. See Figure— 16

(5) Dual Redbird 430. See Figure— 16

(6) Redbird 55 Autopilot. See Figure— 16

(7) Manual and electric pitch trim with indicator. See Figure— 39, 28, 16

Approved Configuration: BE58-XX-S001

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 34

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 37, 38, 19

d. Display Requirements:

(1) This configuration has instruments and

indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 19
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 19

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 19

(4) Redbird 430/530, Transponder, Audio Panel, and Redbird 140 Autopilot. See Figure— 19

(5) Redbird 430/530. See Figure— 19

(6) Redbird 140 Autopilot. See Figure— 19

(7) Manual pitch trim with indicator. See Figure— 39, 19

Approved Configuration: PA34-XX-S001

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 34

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 37, 38, 23

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:

Traditional flight instruments. See Figure— 23
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 23

(3) Life sized traditional “round” gages and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 23

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 23

(5) Redbird 430/530. See Figure— 23

(6) Redbird 140 Autopilot. See Figure— 23

(7) Manual pitch trim with indicator. See Figure— 39, 23

Approved Configuration: PA44-XX-S001

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure—34
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.
This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 24

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:

Traditional flight instruments. See Figure— 24
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 24

(3) Life sized traditional “round” gages and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 24

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 24

(5) Redbird 430/530. See Figure— 24

(6) Redbird 140 Autopilot. See Figure— 24

(7) Manual pitch trim with indicator. See Figure— 39, 24

Approved Configuration: C172-FC-G003

This configuration represents a single engine land, fixed gear, fixed pitch propeller, high wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and an Integrated Redbird 700 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has mixture and throttle controls. See Figure— 30
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 4

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD

with standby instruments. See Figure— 4 Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 4

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 4

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Integrated Redbird 700 Autopilot. See Figure— 4

(5) Glass Cockpit Redbird 1000 PFD & MFD . See Figure— 4

(6) Integrated Redbird 700 Autopilot. See Figure— 4

(7) Manual pitch trim with indicator. See Figure— 39, 4

Approved Configuration: C182-FC-G003

This configuration represents a single engine land, fixed gear, constant speed propeller, high wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and an Integrated Redbird 700 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 31
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 7

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:

Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 7
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass. Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 7

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 7

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Integrated Redbird 700 Autopilot. See Figure— 7

(5) Glass Cockpit Redbird 1000 PFD & MFD. See Figure— 7

(6) Integrated Redbird 700 Autopilot. See Figure— 7

(7) Manual pitch trim with indicator. See Figure— 39, 7

Approved Configuration: PA28A-XX-G001

This configuration represents a single engine land, fixed gear, fixed pitch propeller, low wing, airplane with Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has mixture and throttle controls. See Figure— 32
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 9

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:

Glass Cockpit Redbird Entegra PFD & MFD with standby instruments. See Figure— 9
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Dual Redbird 430)
Navigation radio that meets the requirements listed in this item. (Dual Redbird 430)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird Entegra PFD & MFD representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 9

(3) Life sized Glass Cockpit Redbird Entegra PFD & MFD and dual Redbird 430 GPS and Navigation radios representative of aircraft being simulated See Figure— 9

(4) Dual Redbird 430, Transponder, Audio Panel, and Redbird 55 Autopilot. See Figure— 9

(5) Dual Redbird 430. See Figure— 9

(6) Redbird 55 Autopilot. See Figure— 9

(7) Manual pitch trim with indicator. See Figure— 39, 9

Approved Configuration: PA28B-XX-S001

This configuration represents a single engine land, fixed gear, fixed pitch propeller, low wing, airplane with traditional "round" gauge instruments, dual Redbird 430, and Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has mixture and throttle controls. See Figure— 32
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Dual Redbird 430).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 10

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:

Traditional flight instruments. See Figure— 10
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass. Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gage applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item (Dual Redbird 430).
Navigation radio that meets the requirements listed in this item (Dual Redbird 430).

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 10

(3) Life sized traditional “round” gauges and dual Redbird 430GPS and Navigation radios representative of aircraft being simulated See Figure— 10

(4) Dual Redbird 430, Transponder, Audio Panel, ADF, and Redbird 55 Autopilot. See Figure— 10

(5) Redbird 430/530. See Figure— 10

(6) Redbird 55 Autopilot. See Figure— 10

(7) Manual pitch trim with indicator. See Figure— 39, 10

Approved Configuration: PA28R-XX-G001

This configuration represents a single engine land, retractable gear, constant speed propeller, low wing, airplane with Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 33
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Dual Redbird 430).
Clock and timer.
Gear handle that functions
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 12

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:

Glass Cockpit Redbird Entegra PFD & MFD with standby instruments. See Figure- 12
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Dual Redbird 430)
Navigation radio that meets the requirements listed in this item. (Dual Redbird 430)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird Entegra PFD & MFD representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 12

(3) Life sized Glass Cockpit Redbird Entegra PFD & MFD and dual Redbird 430 GPS and Navigation radios representative of aircraft being simulated See Figure— 12

(4) Dual Redbird 430, Transponder, Audio Panel, and Redbird 55 Autopilot. See Figure— 12

(5) Dual Redbird 430. See Figure— 12

(6) Redbird 55 Autopilot. See Figure— 12

(7) Manual trim with indicator. See Figure— 39, 12

Approved Configuration: DA20-XX-S007

This configuration represents a single engine land, fixed gear, fixed pitch propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430, and Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:
(1)(a) This configuration has a self-centering displacement center stick. See Figure— 29
(1)(d) This configuration has mixture and throttle controls. See Figure— 32
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 13

d. Display Requirements:
(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented: Traditional flight

instruments. See Figure— 13
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430)
Navigation radio that meets the requirements listed in this item. (Redbird 430)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 13

(3) Life sized traditional “round” gages and Redbird 430 GPS and Navigation radio representative of aircraft being simulated See Figure— 13

(4) Redbird 430, Transponder, Audio Panel, and Redbird 55 Autopilot. See Figure— 13

(5) Redbird 430. See Figure— 13

(6) Redbird 55 Autopilot. See Figure— 13

(7) Manual and electric pitch trim with indicator. See Figure— 39, 13

Approved Configuration: DA40-FC-G003

This configuration represents a single engine land, fixed gear, constant speed propeller, low wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and a Integrated Redbird 700 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement center stick. See Figure— 29
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 33
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 14

d. Display Requirements:

(1) This configuration has instruments and

indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 14
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass. Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria
a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 14

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 14

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Integrated Redbird 700 Autopilot. See Figure— 14

(5) Glass Cockpit Redbird 1000 PFD & MFD. See Figure— 14

(6) Integrated Redbird 700 Autopilot. See Figure— 14

(7) Manual and electric pitch trim with indicator. See Figure— 39, 14

Approved Configuration: SR22-FC-G004

This configuration represents a single engine land, fixed gear, constant speed propeller, low wing, airplane with a Glass Cockpit Redbird Perspective PFD & MFD with Redbird FMS Keypad and Redbird Perspective Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement side yoke. See Figure— 28
(1)(d) This configuration has mixture and throttle controls. See Figure— 32
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird Perspective).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 17

d. Display Requirements:

(1) This configuration has instruments and

indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird Perspective PFD & MFD with Redbird FMS Keypad and Redbird Perspective Autopilot with standby instruments. See Figure— 17
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio meets the requirements listed in this item. (Redbird Perspective)
Navigation radio that meets the requirements listed in this item. (Redbird Perspective)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 17

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 17

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Integrated Redbird 700 Autopilot. See Figure— 17

(5) Glass Cockpit Redbird 1000 PFD & MFD. See Figure— 17

(6) Integrated Redbird 700 Autopilot. See Figure— 17

(7) Manual pitch and electric trim with indicator. See Figure— 39, 28, 17

Approved Configuration: BE55-XX-S008

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 35

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 37, 38, 18

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 18
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 18

(3) Life sized traditional “round” gages and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 18

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 18

(5) Redbird 430/530. See Figure— 18

(6) Redbird 140 Autopilot. See Figure— 18

(7) Manual pitch trim with indicator. See Figure— 39, 18

Approved Configuration: BE58-FC-G003

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and an Integrated Redbird 700 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 34
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 20

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 20
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 20

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 20

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Integrated Redbird 700 Autopilot. See Figure— 20

(5) Glass Cockpit Redbird 1000 PFD & MFD. See Figure— 20

(6) Integrated Redbird 700 Autopilot. See Figure— 20

(7) Manual pitch trim with indicator. See Figure— 39, 20

Approved Configuration: BE76-XX-S006

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27
(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 34
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 37, 38, 22

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 22
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 22

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 22

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 22

(5) Redbird 430/530. See Figure— 22

(6) Redbird 140 Autopilot. See Figure— 22

(7) Manual pitch trim with indicator. See Figure— 39, 22

Approved Configuration: PA30-XX-S009

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 34

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 37, 38, 22

d. Display Requirements:

(1) This configuration has instruments and

indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 22
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 22

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated. See Figure— 22

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 22

(5) Redbird 430/530. See Figure— 22

(6) Redbird 140 Autopilot. See Figure— 22

(7) Manual pitch trim with indicator. See Figure— 39, 22

Approved Configuration: DA42-FC-G003

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and an Integrated Redbird 700 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement center stick. See Figure— 29

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 36

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 37, 38, 25

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 25
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 25

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 25

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Integrated Redbird 700 Autopilot. See Figure— 25

(5) Glass Cockpit Redbird 1000 PFD & MFD. See Figure— 25

(6) Integrated Redbird 700 Autopilot. See Figure— 25

(7) Manual pitch trim with indicator. See Figure— 39, 25

Approved Configuration: GA7-XX-S009

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 27

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 34

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 37, 38, 26

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 26
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass. Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 26

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated. See Figure— 26

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 26

(5) Redbird 430/530. See Figure— 26

(6) Redbird 140 Autopilot. See Figure— 26

(7) Manual pitch trim with indicator. See Figure— 39, 26

Section 4: Figures and Photos

Figure 1: Generic Complex Single Engine—Land



Figure 2: C172-XX-S001



Figure 3: C172-XX-G001



Figure 4: C172-FC-G003



Figure 5: C182-XX-S001



Figure 6: C182-XX-G001



Figure 7: C182-FC-G003



Figure 8: PA28A-XX-S001



Figure 9: PA28A-XX-G001



Figure 10: PA28B-XX-S001



Figure 11: PA28R-XX-S001



Figure 12: PA28R-XX-G001



Figure 13: DA20-XX-S007



Figure 14: DA40-FC-G003



Figure 15: SR20-XX-G001



Figure 16: SR22-XX-G001



Figure 17: SR22-FC-G004



Figure 18: BE55-XX- S008



Figure 19: BE58-XX-S001



Figure 20: BE58-FC-G003



Figure 21: BE76-XX-S006



Figure 22: PA30-XX-S009



Figure 23: PA34-XX-S001



Figure 24: PA44-XX- S001



Figure 25: DA42-FC-G003



Figure 26: GA7-XX-S009



Figure 27: Standard Yoke



Figure 28: Side Yoke



Figure 29: Center Stick



Figure 30: Single Engine Vernier Throttle Quadrant



Figure 31: Single Engine Complex Vernier Throttle Quadrant



Figure 32: Single Engine "Boat Style" Throttle Quadrant

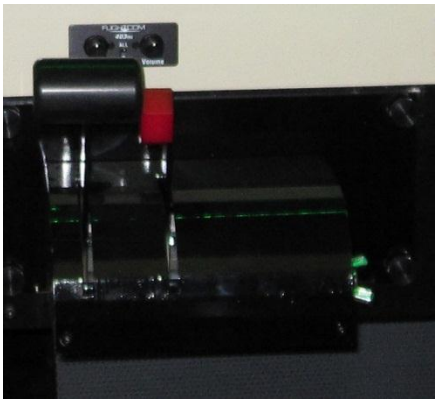


Figure 33: Single Engine Complex "Boat Style" Throttle Quadrant

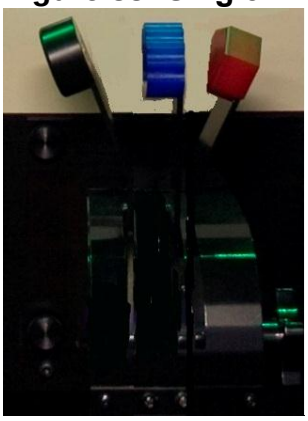


Figure 34: Twin Engine "Boat Style" Throttle Quadrant

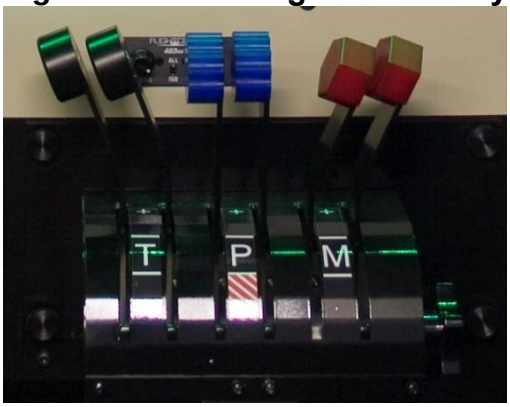


Figure 35: Twin Engine Reverse "Boat Style" Throttle Quadrant

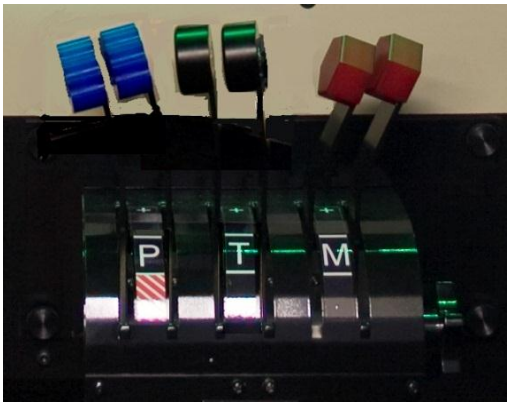


Figure 36: Twin Engine FADEC "Boat Style" Throttle Quadrant

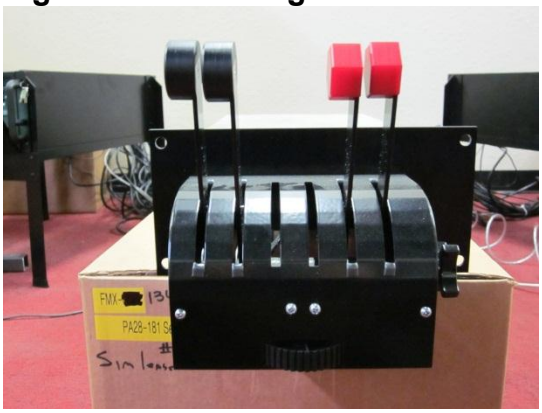


Figure 37: Right Side of Lower Switch Panel



Figure 38: Left Side of Lower Switch Panel



Figure 39: Manual Trim Wheel



Figure 40: Instructors Tablet PC



Figure 41: Training Session Buttons

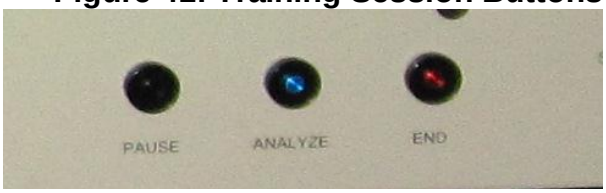


Figure 42: Rudder Pedals

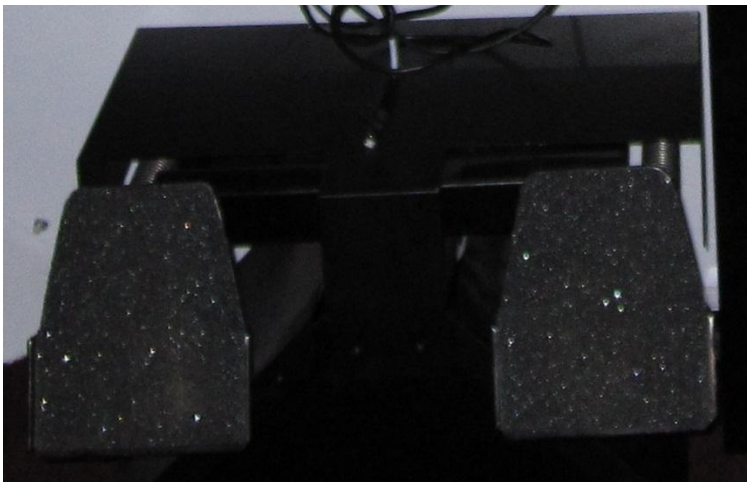


Figure 43: Visual Displays



Figure 44: Assembled Cockpit Configuration



Figure 45: Assembled Cockpit Configuration



Figure 46: External Shroud LD



Figure 47: External Shroud SD



Figure 48: External Shroud FMX


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